

Tutorial 4

ECON 101 – Macroeconomics

Fall 2025

1. The Basic Multiplier in a Closed Economy (No Taxes, No Trade)

Consumption Function:

$$C = \bar{C} + cY$$

where c is the marginal propensity to consume (MPC).

Aggregate Expenditure:

$$AE = C + I + G = \bar{C} + cY + I + G$$

Equilibrium occurs when $Y = AE$.

Multiplier (Closed Economy):

$$k = \frac{1}{1 - c}$$

Example Parameters:

$$\bar{C} = 200, \quad c = 0.75, \quad I = 150, \quad G = 250$$

Step 1: Solve for Equilibrium Income

$$Y = \bar{C} + cY + I + G$$

$$Y - 0.75Y = 200 + 150 + 250$$

$$0.25Y = 600 \quad \Rightarrow \quad Y = \boxed{2,400}$$

Step 2: Apply a Government Spending Increase

$$\Delta G = +100, \quad k = \frac{1}{1 - 0.75} = 4$$

$$\Delta Y = k \cdot \Delta G = 4 \times 100 = \boxed{400}$$

$$Y' = 2,400 + 400 = \boxed{2,800}$$

Step 3: Round-by-Round Spending Illustration

$$100 + 0.75(100) + 0.75^2(100) + \cdots = \frac{100}{1 - 0.75} = 400$$

Each round induces additional consumption until total change in output equals $\Delta Y = 400$.

2. The Multiplier with Taxes and Imports (Open Economy)**Setup:**

$$C = \bar{C} + c(1 - t)Y, \quad NX = \bar{X} - mY$$

$$AE = \bar{C} + I + G + \bar{X} + [c(1 - t) - m]Y$$

Define $A_0 = \bar{C} + I + G + \bar{X}$ (autonomous expenditure), and the slope of the AE function as $z = c(1 - t) - m$.

Equilibrium:

$$Y = A_0 + zY \quad \Rightarrow \quad Y(1 - z) = A_0$$

$$Y = \frac{A_0}{1 - z}$$

Multiplier:

$$k = \frac{1}{1 - [c(1 - t) - m]} = \frac{1}{1 - c(1 - t) + m}$$

Example Parameters:

$$\bar{C} = 200, \quad c = 0.75, \quad I = 150, \quad G = 250, \quad \bar{X} = 20, \quad t = 0.20, \quad m = 0.15$$

Step 1: Compute the Slope z

$$z = c(1 - t) - m = 0.75(1 - 0.20) - 0.15 = 0.75(0.80) - 0.15 = 0.60 - 0.15 = \boxed{0.45}$$

Step 2: Compute Autonomous Spending A_0

$$A_0 = \bar{C} + I + G + \bar{X} = 200 + 150 + 250 + 20 = \boxed{620}$$

Step 3: Find Equilibrium Income

$$Y = \frac{A_0}{1 - z} = \frac{620}{1 - 0.45} = \frac{620}{0.55} = \boxed{1,127.27}$$

Step 4: Compute the Multiplier

$$k = \frac{1}{1 - c(1 - t) + m} = \frac{1}{1 - 0.75(0.80) + 0.15}$$

$$k = \frac{1}{1 - 0.60 + 0.15} = \frac{1}{0.55} = \boxed{1.8182}$$

Step 5: Fiscal Policy Shock

$$\Delta G = +100$$

$$\Delta Y = k \cdot \Delta G = 1.8182 \times 100 = \boxed{181.82}$$

$$Y' = 1,127.27 + 181.82 = \boxed{1,309.09}$$

Step 6: Spending Rounds Intuition (with Leakages) Each spending round induces a smaller response:

$$100 + 0.45(100) + 0.45^2(100) + 0.45^3(100) + \dots = \frac{100}{1 - 0.45} = 181.82$$

Interpretation: Taxes ($t = 0.20$) and imports ($m = 0.15$) create leakages that reduce the multiplier from 4.0 in a closed economy to approximately 1.82. This illustrates why fiscal policy has a smaller impact in open economies with taxation and trade.

End of Example